ShadowNET: Novel RGB-D Ensemble Model for Object Classification in Uncontrolled Light Settings

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Summary

- Used transfer learning to train a ensemble model on RGB-D data covering only partial object views.
- Created a custom dataset of geometric objects in uncontrolled light settings,
- Comparing the performance of RGB-D v. RGB models w/ bad lighting data

Background

- It is unclear which RGB-D architectures yields maximal performance advantage over standard RGB models for object classification.
- For practical usage, it may be useful to explore partial-view depth maps rather than complete 3D reconstructions.

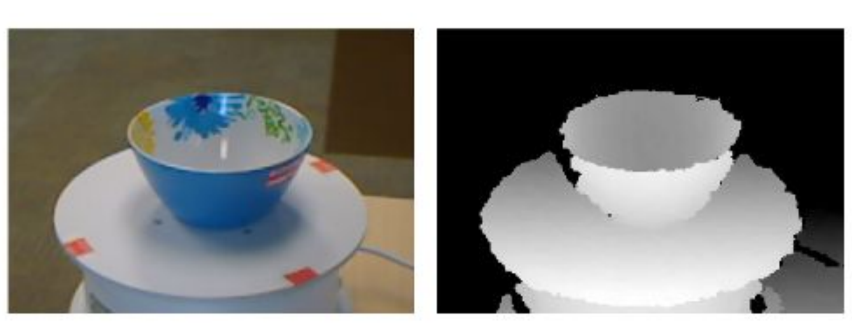
RGB Image   Depth Map

- The utility of depth data alongside color images for classifying objects in uncontrolled lighting settings (shadows, saturation, luminosity) is yet to be explored.

Questions

Can transfer learning be applied to RGB deep learning models to effectively create an RGB-D architecture? To what extent is there a benefit to RGB-D over RGB? In uncontrolled light settings? In what situations does RGB-D architecture perform better than RGB models, and vice versa?

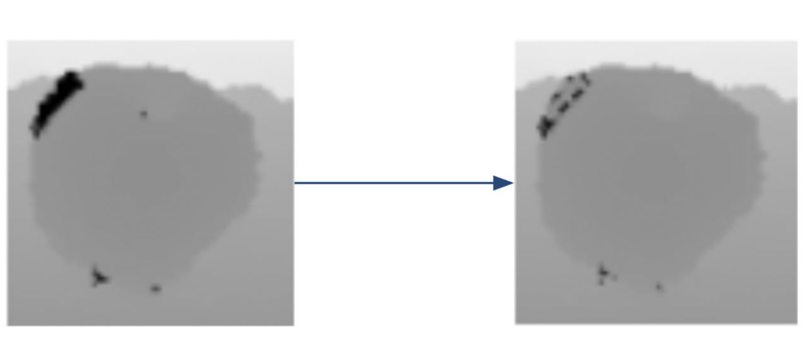
UW RGB-D Object Dataset (ROD)[1]



- 51 objects, 300 classes
  - Fixed pose RGB-D scans taken as objects rotated over 2π radians
  - Concatenated to $2\pi/3$ radians to limit partial-view

Adaptive Median Filtering (AMF)

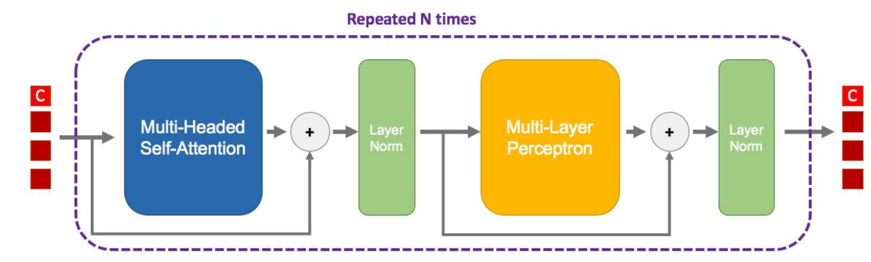
- NaN-depth values cleaned using dynamic sliding window algorithm



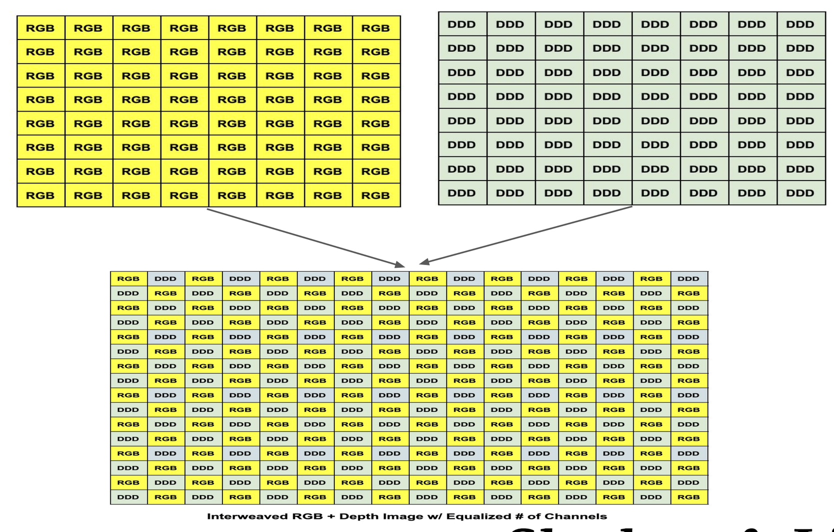
Raw Depth   Filtered

Vision Transformer Model (ViT)

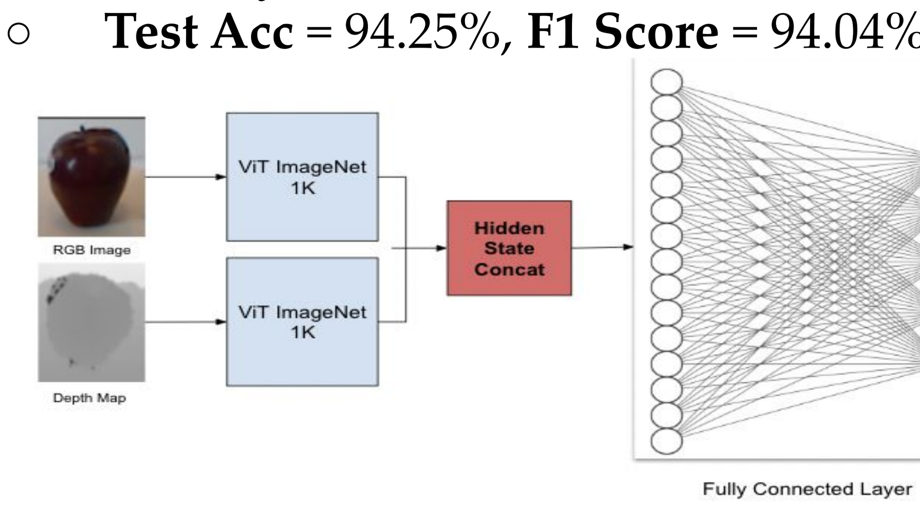
Vision Transformers (ViT)[2] utilize a self-attention mechanism and token positional embeddings to find complex dependencies between image patches. We utilize transfer learning from a ViT trained on ImageNet RGB images for RGB-D:



- **Interleaving (I-ViT)**: pixels from the color images are interleaved with the depth map to form a input lattice of color and depth pixels.
  - **Test Acc** = 91.3%, **F1 Score** =



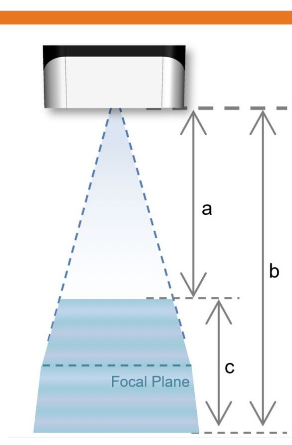
- **Late-Fusion (LF-ViT)**: Color image and depth map passed separately to two ViT structures; final hidden states are concatenated and classified in fully connected layer
  - **Test Acc** = 94.25%, **F1 Score** = 94.04%



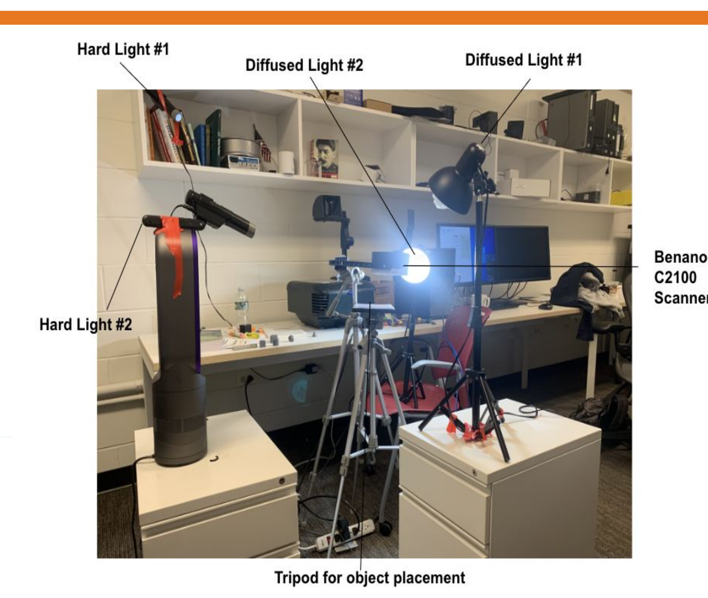
Shadow & Lighting Dataset Creation

We offer the Princeton Geometric Dataset (PGD)

- 7 gray geometric objects subject to bad lighting
- Shadow (S) & non-shadow (NS) scans per object
  - Shadows: 4 lights are cast on object
  - Non-shadow: 0 lights cast on object
- Camera (Benano C2100 scanner) initially at 0^{p} pose angle. Each geometric object is rotated 45° after each scan. The camera pose increases in incline angle per revolution (0°, 5°, 15°, 25°)



Projected Light Scanning[3]



Dataset Collection Setup, Lateral View

Experiment: LF-ViT vs. Flir Firefly DL Camera[4]

We finetune LF-ViT on PGD and compare it to the Flir Firefly DL Camera (FLIR), a commercially available DL-inference camera, using FLIR as a baseline over multiple train/test splits.

Firefly FLIR DL Camera (FC)

- Only has color capabilities; could help show that without depth, an RGB-NN may be unable to classify images with multiple shadows when trained on images with no shadows, and vice versa.
- Mobile Net v24 1.0 architecture, RGB-CNN model.
- Camera network trained from scratch using Movidius chip. Tested DL-inference w/ PGD using injection through modifying associated scripts from Spinnaker

Training Specs:

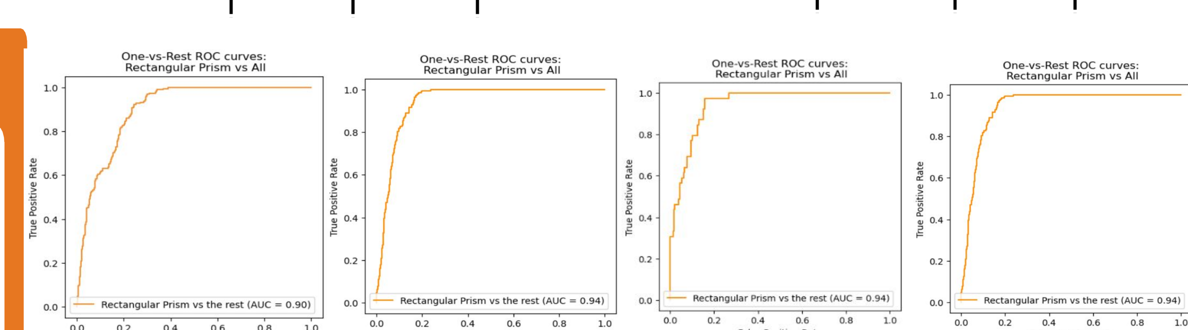
- 4 training splits: Train NS, Test S; Train S, Test NS; Train S, Test S; Train NS, Test NS
- Used image translation/rotation to augment sets (~1350 shadow images, 1344 nonshadow images)

	LF-ViT	FLIR	LF-ViT	FLIR
Train, Test	Accuracy	Accuracy	F1	F1
NS, S	0.81	0.78	0.79	0.48
S, NS	0.82	0.79	0.79	0.58
S, S	0.99-1.0	0.97-1.0	0.94	-
NS, NS	0.99-1.0	0.98-1.0	0.96	-

Train NS, Test S				Train S, Test NS			
Object	**Correct (LF-ViT)**	**Correct (FLIR)**	**Total**	**Object**	**Correct (LF-ViT)**	**Correct (FLIR)**	**Total**
Cube	**30**	**125**	**192**	**Cube**	**17**	**70**	**192**
Cylinder	**192**	**137**	**192**	**Cylinder**	**192**	**191**	**192**
Hemisphere	**157**	**160**	**198**	**Hemisphere**	**177**	**192**	**192**
Hex. Prism	**176**	**186**	**192**	**Hex. Prism**	**192**	**192**	**192**
Rect. Prism	**169**	**67**	**192**	**Rect. Prism**	**179**	**114**	**192**
Sphere	**178**	**191**	**192**	**Sphere**	**179**	**106**	**192**
TPrism	**190**	**182**	**192**	**TPrism**	**168**	**192**	**192**
Accuracy	**1092/1350 (0.81)**	**1048/1350 (0.78)**		**Accuracy**	**1104/1344 (0.82)**	**1057/1344 (0.79)**	

Discussion

- AMF fills gaps/NaNs from depth map data
- On ROD, I-ViT (91.3%) performs worse than w/ standard RGB-ViT (93%)
- Transfer w/ fusion = better performance than standard RGB-ViT on ROD/PGD. Less parameters & using augmentation reduces overfitting. LF-ViT performs better on ROD (94.25%) than a CNN RGB-D paradigm (87.8%)
- LF-ViT demonstrates improved plasticity between lighting settings (shadow v. non-shadow) than FLIR
- LF-ViT confounds similar object paradigms (cyl-sph-hemi, cube-rect) lesser than FLIR but worse on hex/tri. prisms
- LF-ViT proves viability of public grassroots image dataset creation w/ depth scan technologies in phones
- Possible future modifications of LF-ViT for medical segmentation applications (i.e. tumors and wounds)



References & Acknowledgements

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Princeton Geometric Dataset: **tinyurl.com/princetongd**

[1] K. Lai, L. Bo, X. Ren and D. Fox, "A large-scale hierarchical multi-view RGB-D object dataset," 2011 IEEE International Conference on Robotics and Automation
[2] Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).
[3] "C2100." Benano Inc., www.benanos.com/product?id=23
[4] Firefly DL | Teledyne FLIR, www.flir.com/products/firefly-dl